

Final Project: Network Design and Security Configuration

Estimated time needed: **60** minutes

Final project requirements:

You will use an online Windows environment for this project.

Required Tools and Resources:

- A screenshot or screen-capturing application, with the ability to crop images
- A location to store your screenshots

Optional: Using Your Own Machine If you choose to complete this project on your own computer, you will need:

- A workstation running Microsoft Windows 10 or later
- Administrator rights on the workstation

Project overview

This final project consists of four real-world, network-related tasks. Some tasks may include several subtasks. For each task, you will capture and save screenshots that document your successful completion of the required steps. You will later submit these screenshots for evaluation as part of your course completion.

Let's get started!

Scenario:

TechSafe, Ltd. has hired you to design and secure its network. The company uses a fixed IP address from its Internet service provider (ISP) and wants a structured network with compatible IP addressing, subnetting, and security configurations.

Task 1: Design a network with subnets

- **Scenario:** Design a network diagram for TechSafe Ltd that includes three departments: Administration, Sales, and IT. Each department requires its own subnet.
- **Hints**

► Hint 1:

► Hint 2

► Hint 3:

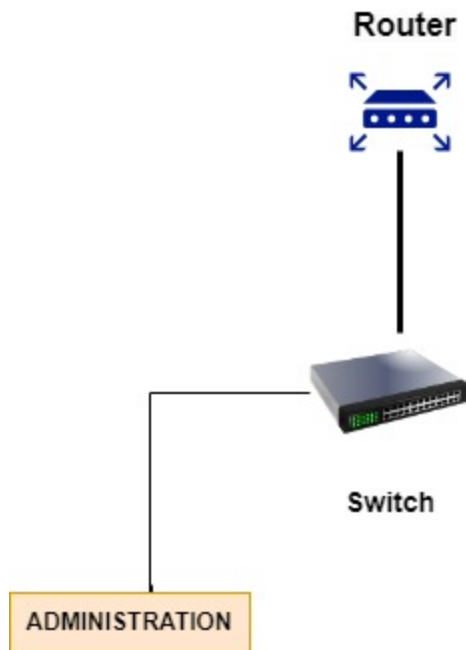
- **Deliverable:** A screenshot of the network diagram.

The submitted network screenshot must include the following devices and labels:

- **One router** connected to **one switch**. The router connects to the switch with a line. Label the router as **Router** and label the switch as **Switch**.
- **Three subnets** that connect to the switch using solid lines The three subnets are labeled as **Administration**, **Sales**, and **IT**.
- The diagram includes no additional elements beyond the router, switch, and subnets.

Name the screenshot **Task1NetworkDiagram** and save it in either PNG or JPG format in a location where you can easily upload it for submission.

- **Sample Output**



Note: This image is a sample and might not include all of the assignment requirements.

Task 2: Calculate Subnet Addresses and Subnet Masks

- **Scenario:** You are working with a fixed IP address of 198.51.100.0/24. The network is divided into three subnets with the following IP address ranges for each department:
 - **Administration:** 198.51.100.1 - 198.51.100.62
 - **Sales:** 198.51.100.65 - 198.51.100.126
 - **IT:** 198.51.100.129 - 198.51.100.190

Your task is to calculate each department's subnet addresses and subnet masks based on their specified IP address ranges and save that information into a text-based table for later use.

- **Hints**

► **Hint 1**

► **Hint 2**

► **Hint 3**

- **Deliverable:** A screenshot of the department subnet addresses, masks, and address ranges Before capturing the screenshot, verify that your table displays the following information:
 - Four columns: Subnet Name, Subnet Address, Subnet Mask, and IP Address Range. These labels are displayed at the top of each column.
 - One row for each subnet (Administration, Sales, IT).
 - Each subnet name's subnet address, mask, and IP address range.
 - No additional text beyond the table.

Next, capture your screenshot and name the screenshot **Task2Screenshot1** and save it in either PNG or JPG format in a location where you can easily upload it for submission.

- **Sample Output**

Subnet Name	Subnet Address	Subnet Mask	IP Address Range
Administration	xxx.xxx.xxx.xxx	xxx.xxx.xxx.xxx	198.51.100.1 - 198.51.100.62
Sales	xxx.xxx.xxx.xxx	xxx.xxx.xxx.xxx	198.51.100.65 - 198.51.100.126
IT	xxx.xxx.xxx.xxx	xxx.xxx.xxx.xxx	198.51.100.129 - 198.51.100.190

Task 3: Analyze HTTP/HTTPS Website Traffic for Using Wireshark

- **Scenario:** After implementing the network design in Task 1 and configuring the IP addresses and subnets during Task 2, TechSafe Ltd. needs you to monitor network usage related to website that their employees might access. Your task is to capture and analyze HTTP/HTTPS traffic from one of the following websites to identify and understand the data flow related to these two services.

Hints: Apply the following steps to perform your analysis:

1. Use Wireshark to capture live network traffic on your system while you visit Wikipedia and Google. Spend at least 2 minutes actively browsing both sites.
2. Apply the following filters in Wireshark:
 - Since these websites use **HTTPS**, filter the captured traffic for **HTTPS** by applying a filter for tcp port **443**.
 - Add additional filter conditions to search for tcp containing these websites to isolate packets related to these domains.
 - In this example, let's use the examples of Wikipedia and Google. The final filter will look like this: **tcp.port==443 and tcp contains "Wikipedia"** or **tcp.port==443 and tcp contains "Google"**.
 - Analyze the filtered traffic to identify the domains, URLs, or types of content being accessed on Wikipedia and Google.

Select a website from the following list for this task.

Amazon.com

Office.com

Ebay.com

Sharepoint.com

Google.com

Weather.com

LinkedIn.com

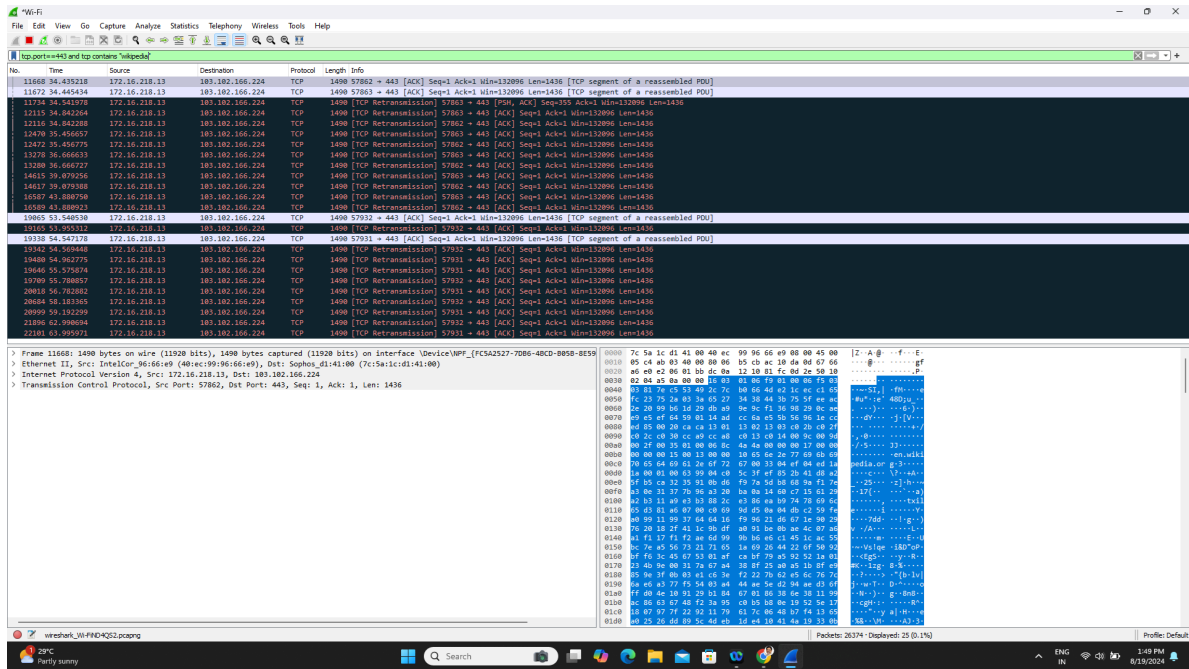
Whatapp.com

Live.com

Wikipedia.com

Deliverables: Submit a screenshot of the Wireshark application displaying filtered

• Sample Output



Note: This sample output is from different site from which you will collect your data. This screenshot does not include all of the data you will submit with your screen capture.

Task 4: Configure Firewall Rules Using Microsoft Windows Defender Firewall

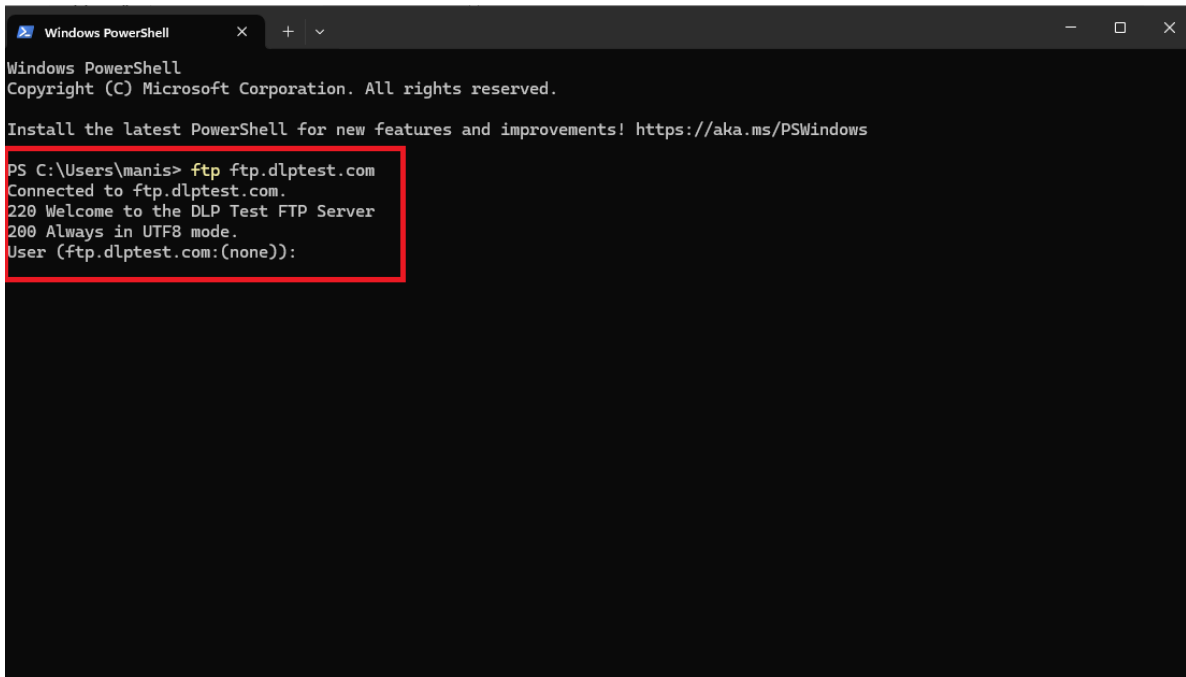
Scenario: TechSafe Ltd. wants to further secure its internal network by restricting access to specific types of web traffic. To enhance security, you must go to each workstation and configure Microsoft Windows Defender Firewall to block all FTP and HTTP traffic.

Hints:

Step 1: Open a terminal window and type `ftp ftp.dlptest.com` Capture a screenshot that shows successful access to `ftp ftp.dlptest.com` before applying the firewall rule.

Deliverable: Name the screenshot **Task4Screenshot1** and save the screenshot in an

- **Sample Output:** Screenshot of access to FTP before applying the firewall rule



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\manis> ftp ftp.dlptest.com
Connected to ftp.dlptest.com.
220 Welcome to the DLP Test FTP Server
200 Always in UTF8 mode.
User (ftp.dlptest.com:(none)):
```

Step 2: Open **Microsoft Windows Defender Firewall** with Advanced Security. Begin creating a new outbound rule to block FTP traffic by selecting Port as the rule type. Choose the TCP protocol and enter 21 as the Remote Port. Capture a screenshot of these settings.

Deliverable: Name the screenshot **Task4Screenshot2** and save it in an accessible location for later uploading.

Step 3: Verify that the rule to block FTP access is working. Open a terminal window and type `ftp ftp.dlptest.com` to test FTP connectivity. Capture a screenshot showing the attempt to access `ftp.dlptest.com` after the rule has been applied.

Deliverable: Name the screenshot **Task4Screenshot3** and save the screenshot in an accessible location for later uploading.

Step 4: Open a browser and access <http://httpforever.com/>. If you see a cautionary message that this website is not secured, select the onscreen message that enables you to proceed to the site.

Deliverable: Name the screenshot as **Task4Screenshot4** and save the screenshot in an accessible location for later uploading.

Step 5: Open **Microsoft Windows Defender Firewall** with Advanced Security. Begin

Choose the TCP protocol and enter 80 as the Remote Port. Capture a screenshot of these settings.

Deliverable: Name the screenshot **Task4Screenshot5** and save the screenshot in an accessible location for later uploading.

Step 6: Verify that the HTTP traffic is active and correctly configured within your firewall settings by attempting to access <http://httpforever.com/>. Capture a screenshot that shows the attempt to access <http://httpforever.com/> after applying the firewall rule, confirming that access is blocked.

Deliverable: Name the screenshot **Task4Screenshot6** and save the screenshot in an accessible location for later uploading.

Conclusion

Congratulations! You did it! This project provided a comprehensive experience in network design and security configuration. By creating a structured network with proper IP allocation and subnetting, you ensured that TechSafe Ltd.'s network is both organized and efficient. Your network traffic analysis using Wireshark demonstrated your understanding of how specific services like Wikipedia and Google interact with network protocols, while configuring firewall rules using Microsoft Windows Defender enhanced the security posture of the network. Overall, this project demonstrated your ability to apply critical networking concepts and security measures in a real-world scenario, ensuring that a company's infrastructure is both robust and secure.

Author(s)

Dr. Manish Kumar

Other Contributor(s)

Patsy Kravitz

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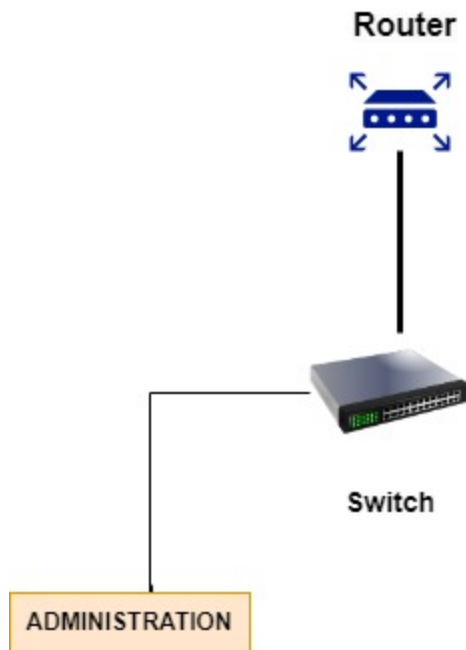
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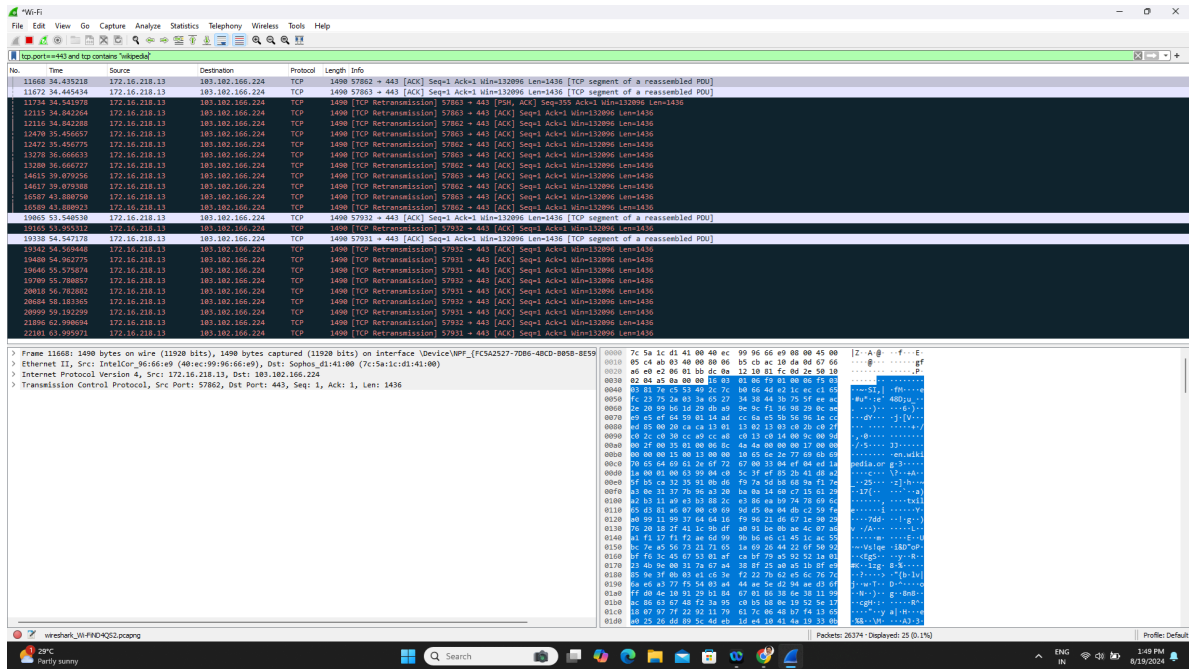
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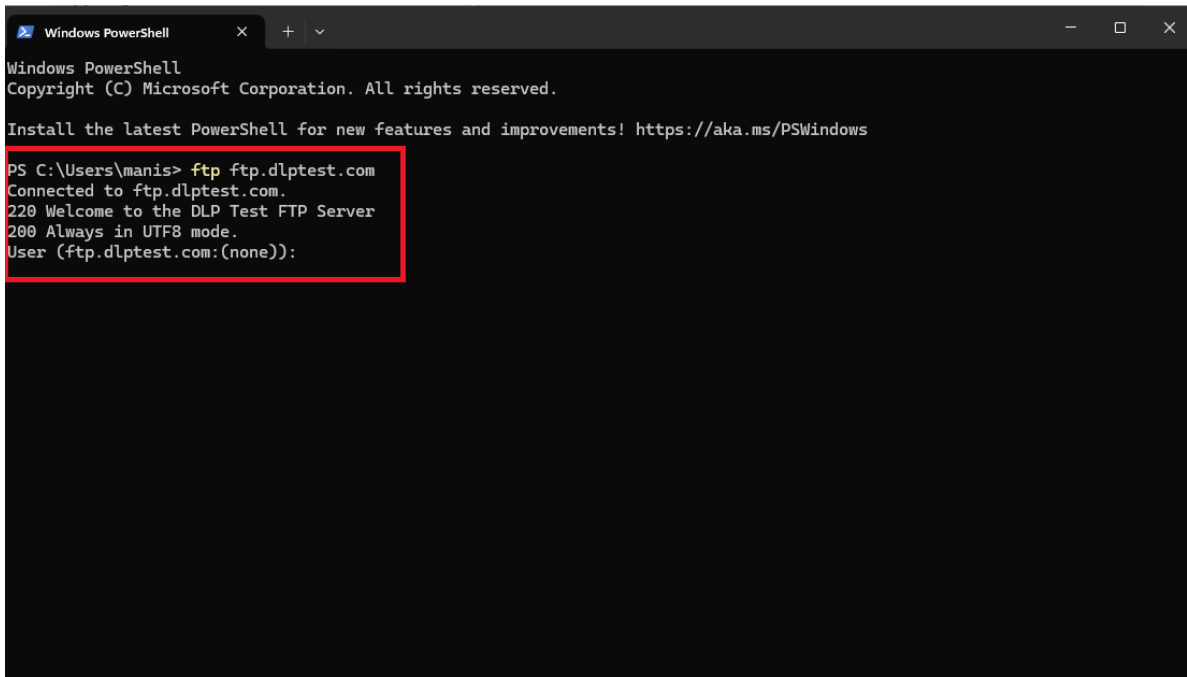
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Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

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Deliverable: Name the screenshot **Task4Screenshot3** and save the screenshot in an accessible location for later uploading.

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Deliverable: Name the screenshot as **Task4Screenshot4** and save the screenshot in an accessible location for later uploading.

Step 5: Open **Microsoft Windows Defender Firewall** with Advanced Security. Begin

Choose the TCP protocol and enter 80 as the Remote Port. Capture a screenshot of these settings.

Deliverable: Name the screenshot **Task4Screenshot5** and save the screenshot in an accessible location for later uploading.

Step 6: Verify that the HTTP traffic is active and correctly configured within your firewall settings by attempting to access <http://httpforever.com/>. Capture a screenshot that shows the attempt to access <http://httpforever.com/> after applying the firewall rule, confirming that access is blocked.

Deliverable: Name the screenshot **Task4Screenshot6** and save the screenshot in an accessible location for later uploading.

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